

QUANTITATIVE CHEMISTRY – REACTING MASSES

KEY VOCABULARY

Stoichiometry: the ratio of reactants and products in an equation.

Limiting reactant: the reactant that runs out first and stops the reaction.

Excess: a reactant that is not all used up.

Mole ratio: the ratio of moles from the balanced equation.

COMMON MISCONCEPTIONS

~~—The reactant with the bigger mass is in excess.~~

✓ Compare MOLES, not masses, to find excess.

~~—Mass ratio = mole ratio.~~

✓ Convert to moles first using the molar mass.

~~—All reactants are used up.~~

✓ Often one is in excess and is left over.

WORKED EXAMPLE

$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. Find MgO from 48 g Mg.

moles Mg = $48 \div 24 = 2 \text{ mol}$

ratio Mg : MgO = 2 : 2 $\rightarrow$ 2 mol MgO

mass MgO = 2×40

= 80 g

1 USE THE RATIO

$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. Calculate the mass of water from 4 g of H_2 . ($M_r \text{H}_2=2$, $\text{H}_2\text{O}=18$) [3 marks]

Answer: _____ g

2 FIND THE REACTANT

$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$. Calculate the mass of CaCO_3 needed to make 5.6 g CaO. ($\text{CaCO}_3=100$, $\text{CaO}=56$) [3 marks]

Answer: _____ g

3 LIMITING REACTANT

Explain what is meant by the limiting reactant. [1 mark]

4 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Convert masses to _____ first. The reactant that runs out is the _____ reactant.

moles

limiting

mass

excess

5 IDENTIFY EXCESS

Explain how you would decide which reactant is in excess. [2 marks]

6 SCALE UP

If 24 g of Mg makes 40 g MgO, calculate the MgO from 12 g of Mg. [2 marks]

Answer: _____ g